

Visual features

Spectrogram image
 $224 \times 224 \times 3$ (RGB)

EfficientNet-B0 backbone (ImageNet pretrained)

Conv Stem
Initial convolution

MBConv Blocks
8 stages, expansion

Feature Map
 $B \times 1280 \times 7 \times 7$

Attention

Temporal Reshape

Pool $\rightarrow$ squeeze $\rightarrow$ permute $\rightarrow B \times 7 \times 1280$

Linear Projection

$1280 \rightarrow 256$ ($B \times 7 \times 256$)

Temporal Attention — Layer 1

Head 1
 $Q \cdot K^T \rightarrow V$

Head 2
 $Q \cdot K^T \rightarrow V$

Head 3
 $Q \cdot K^T \rightarrow V$

Head 4
 $Q \cdot K^T \rightarrow V$

Concat + Add & LayerNorm

Temporal Attention — Layer 2

Same structure — $B \times 7 \times 256$

Aggregation

Temporal Mean Pooling

$\text{mean}(\text{dim}=1) \rightarrow B \times 256$

Classifier head

Linear $256 \rightarrow 128$ — GELU + Drop — Linear $128 \rightarrow N$ classes

Genre score (logit)